Sofiane CHALAL

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Bio

My research covers an interdisciplinary topic between probability theory, quantum physics and control theory. My work deals with observed quantum particle systems and their mean-field graph limits.

Education

CentraleSupélec | Paris-Saclay *PHD* : Applied mathematics 2026

Dauphine | PSL *Master* : Applied and Theoretical Mathematics - Probability 2021

Research

Journal articles & preprints

Quantitative propagation of chaos for Lindblad dynamics 2026

with : Nina H. Amini | Journal of Statistical Physics

On the mean-field Belavkin filtering equation 2023

with : Nina H. Amini, Gaoyue Guo | IEEE Control Systems Letters

On brownian motion on the Fubini extension space 2025

with : Hamed Amini, Nina H. Amini, Gaoyue Guo | Submitted to Stochastic Processes and their Applications

Graphon quantum filtering system 2025

with : Hamed Amini, Nina H. Amini, Gaoyue Guo |Submitted to Annales Henri Poincaré - A

Semi-Classical limits for Belavkin filtering equation 2026

with : Nina H. Amini, | In preparation

Mean-field limit for Hudson-Parthasarathy equation 2026

with : Nina H. Amini | In preparation

Mean-field Monte-Carlo approach for N-body open quantum system 2026

with : Nina H. Amini | In preparation

On modeling quantum differential and mean-field games 2026

with : Hamed Amini, Nina H. Amini, Mathieu Laurière | In preparation

Conference proceedings

Dynamic programming principle and stabilization for mean-field quantum filtering systems 2026

with : Hamed Amini, Nina H. Amini, Mathieu Laurière | IEEE qCCL Aalborg

Quantum approach for electrical system maintenance scheduling problem 2026

with : Titouan Carette, Fikri Hafid, Noureddine Henka, Sami Tazi, Djahou Tognon, Qinyan Zhou | ESAIM: Proceedings & Surveys

Unravelling mean-field Lindblad equation 2025

with : Nina H. Amini | IEEE qCCL Hong-Kong

Observed quantum particles system with graphon interaction 2024

with : Hamed Amini, Nina H. Amini, Gaoyue Guo | 63rd IEEE CDC

Teaching and Missions

TD : Probabilités numériques Semester 2 - 2025/2026
Third year, ESILV, 32h

Project supervison : Quantum annealing for portfolio optimization Semester 1 - 2025/2026
Third year, CentraleSupelec, 10h

TD : Statistiques & Outils mathématiques Semester 2 - 2024/2025
First year, IUT Université Paris-Saclay, 32h

Project supervison : QUBO and Quantum Annealing Semester 1 - 2024/2025
Third year, CentraleSupélec, 10h

TD : Equations aux dérivées partielles Semester 1 - 2023/2024
First year, CentraleSupélec, 12h

TD : Integration, Convergence et Probabilités Semester 1 - 2023/2024
First year, CentraleSupélec, 12h

Research mission : Quantum annealing and optimization Semester 2 - 2022/2023
RTE France, 64h

Visit

ICTS - Bengaluru, INDIA January - February 2025

Quantum Trajectories.

IMSI - The University of Chicago, USA November - December 2024

Statistical Methods and Mathematical Analysis for Quantum Information Science.

Invited talks

QP47 Quantum Probability and Infinite Dimensional Analysis - Monastir, TUNISIA October 2026
Open Quantum Klimontovich equation and Empirical Measure in Heisenberg Picture

2nd IEEE Conference on Quantum Control, Computing and Learning - Aalborg, DENMARK July 2026
Dynamic programming principle and stabilization for mean-field quantum filtering systems

Worskshop RTE Chair at CentraleSupelec - Saclay, FRANCE June 2026

Quantum approach for electrical system maintenance scheduling problem

2nd Dolomites WS, Mean-field systems in finance, neurosciences and AI - Folgarida, ITALY January 2026
Observed quantum particles system with mean-field graphon interaction

IEEE Conference on Quantum Control, Computing and Learning - Hong-Kong June 2025
Unravelling mean-field Lindblad equation

63rd IEEE Conference on Decision and Control - Milan, ITALY December 2024

Observed quantum particles system with graphon interaction

26th International Symposium on Mathematical Theory of Networks and Systems - Cambridge, UK August 2024

Control and stabilization of mean-field Belavkin equation

Séminaire d'Automatique du Plateau de Saclay - Saclay, FRANCE February 2024

Quantum mean-field filtering and control

62nd IEEE Conference on Decision and Control - Singapore December 2023

On the mean-field filtering Belavkin equation

GE2MI conference - Levico, ITALY October 2023

Mean-field Belavkin equation and quantum feedback control

Worskshop RTE Chair at CentraleSupelec - Saclay, FRANCE May 2023

Quantum annealing approach to electrical grids problems

Prizes

Best student paper IEEE Conference on Quantum Control, Computing and Learning - Aalborg, 2026.

Skills & Interests

IT : Python (Qutip, Qiskit)
Mathematics : Stochastic Calculus, Stochastic filtering, Monte-Carlo Methods
Interests : I am interested in and enjoy discussing the foundations of quantum mechanics, in particular the measurement problem and the different approaches proposed to address it. For now, I am more convinced by the *eventum mechanics* approach, while remaining open to other viewpoints.